

WORK EXPERIENCE

SENIOR RESEARCHER - ROBOT LEARNING

Mar 2026 - present



Abu Dhabi (UAE) Technology Innovation Institute: Humanoids Group

Developed a real-time Human-Robot Interaction stack for humanoid robots: spoken conversation, event triggering, custom wake-word models, and scene perception, description and question answering, cutting latency 10x. Designed a speech-synchronized gesture server that blends and chains, in real time, kinesthetically taught gestures via Laplacian trajectory editing.

RESEARCHER - AI AND ROBOTICS

Apr 2024 - Dec 2025



Rome (IT) Fucina Nexus Foundation (Remote)

Developed software for a social robot for elderly people and children: natural conversation, detection of anomalies and notable events, and real-time alerts to a custom dashboard.

RESEARCHER - COGNITIVE ROBOTICS AND MANIPULATION

Oct 2021 - Feb 2024



Istanbul (TR) Boğaziçi University: Cognitive Learning and Robotics Laboratory

Enabled industrial and collaborative robots to learn manipulative actions from few-shot demonstrations, adapt and perform different primitives on real-life objects, assess the results to plan and achieve long-term goals.

RESEARCHER - VIRTUAL REALITY

Jun 2022 - Dec 2022



Istanbul (TR) Boğaziçi University Virtual Augmented Reality Laboratory

Integrated particle-based physics for fluids, cloth and deformable objects in Unity VR. Built and animated 3D assets via LiDAR scanning, photogrammetry, mesh cleanup and re-texturing.

RESEARCH ASSISTANT - INTERACTION ROBOTICS

Apr 2020 - Sep 2020



Stockholm (SE) KTH Royal Institute of Technology: Robotic Perception and Learning Department

Developed natural-language interaction software for a humanoid robot: dialogue engine, multi-turn conversation handling, and persistent per-user memory to personalize interactions.

RESEARCH SCHOLAR - COMPUTER VISION

Sep 2018 - Sep 2019



Lisbon (PT) IST Instituto Superior Técnico: I.S.R. Institute for Systems and Robotics - VisLab

Built the perception and navigation stack of a mobile robot: visual SLAM and autonomous indoor navigation, Bluetooth-beacon localization, face detection, object recognition, and an NLP module for voice interaction and parametrized task execution.

PUBLICATIONS

Novel Robotic Action Synthesis with Conditional Neural Movement Primitives.

Lirussi, I. *MSc thesis*, Computer Science and Engineering, University of Bologna, Dec 2023. Supervisor: Prof. Emre Ugur (Boğaziçi University) [PDF]

Framework for Navigation and Human-Robot Interaction with Humanoid Robots.

Lirussi, I. *BSc thesis*, Computer Science and Engineering, University of Bologna, Sep 2019. Supervisor: Prof. Alexandre Bernardino. [PDF]

EDUCATION



MSc. COMPUTER SCIENCE AND ENGINEERING, *University of Bologna* Sep 2019 - Dec 2023

Machine Learning, Distributed Systems, Computational Models, Information Systems, Concurrent and Distributed Programming, Web Services and Applications, Smart City, Development Paradigms, Agile, Scrum, Continuous Integration and Delivery, Launch a Software Startup.



INTERNATIONAL STUDENT MSc., *Stockholm University*

Jan 2020 - Jul 2020

Decision Making and Business Intelligence, Network Security, Cyber Forensics



INTERNATIONAL STUDENT BSc., *Lisbon University*

Sep 2018 - Sep 2019

Artificial Intelligence, Operational Research



BSc. COMPUTER SCIENCE AND ENGINEERING, *University of Bologna* Sep 2014 - Sep 2019

Software Engineering, Embedded Systems and IoT, Automatic Controls, Mobile Apps, Operating Systems, Object-Oriented Programming, Network and Telecommunications, Law for IT, Image Processing, Databases, Algorithms and Data Structures, Computer Architecture, C Programming

TECHNICAL SKILLS & SOFTWARE

- Programming: Python, C++, C, C#, Java, Bash, JavaScript, SQL; OOP, functional and concurrent paradigms.
 - Robotics: ROS/ROS 2 on humanoids, Baxter and UR10; learning from demonstration (CNMP), trajectory generation and editing, visual SLAM (ORB-SLAM) for manipulation; MuJoCo simulation.
 - Motion capture and teleoperation: IMU suits (Xsens), optical marker-based and markerless tracking, Apple Vision Pro teleoperation, motion retargeting.
 - AI: PyTorch, LLM/VLM integration, real-time speech (OpenAI and Gemini Realtime APIs), wake-word models, object detection with OpenCV.
 - VR and 3D: Unity VR development, photogrammetry, LiDAR scanning, mesh cleanup.
 - Deployment and tools: Docker, Linux/WSL, SSH, NVIDIA Jetson (AGX Orin, Thor), FastAPI/Uvicorn, Git/GitLab CI/CD, Jira, Confluence.
- ROS / ROS 2
 - PyTorch
 - Docker
 - OpenCV
 - Git/GitLab/GitHub CI
 - MuJoCo
 - Unity VR
 - ESP32/Arduino/Raspberry
 - LaTeX Writing

LANGUAGES & INTERESTS



IT	Native Speaker
EN	Full Proficiency
PT	Conversational
SV	Basic level
TR	Basic level

Winner "The 30 new emerging ideas of 2021" - Startup Day Unibo
Theoretical and Practical course of First Aid for companies
Cooking and DIY Enthusiast
Sport and Videomaking Passionate
Food Aid Program Volunteer - Blood Donor